

Test And Measurement That's Making AVIATION Safer

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This article discusses the latest test and measurement technologies in the aviation industry

For public safety and the success of military missions, civil aviation and military operations require accurate location, distance and direction of measuring systems. Failure of any of these systems may place human lives at risk. Air navigation systems such as landing systems or *enroute* navigation systems require unique test and measurement (T&M) capabilities.

Innovative T&M solutions allow engineers to identify and fix problems early in the development of a new technology. Oftentimes, waiting until the end of the product lifecycle can prove to be cost-prohibitive. By investing in the correct test solutions, they can reduce cost and, most importantly, risk to projects. Test and verification of avionics hardware and software systems ensure the safety and efficiency of traffic management/air traffic control infrastructures worldwide.

The instrument landing system indicates the distance to the reference point of landing using a combination of radio signals. It provides the aircraft with horizontal and vertical guidance before and during landing. The system also provides

aircraft pilots with landing approach data relative to the ideal landing course. Marker beacon receivers provide signalling output to identify one of three marker beacons installed near the runway and decode audio data.

The lightweight construction of vehicle and machinery components is validated through durability and fatigue testing of materials and structures. These tests include strain gauges, data acquisition systems and software, and force and displacement transducers. Fatigue life of the test specimen is determined by long-term dynamic load testing. To determine maximum static force or stress that the test specimen can tolerate, durability static load testing is done. This includes torsion, tensile, flexure and compression tests.

Some instruments

Madhukar Tripathi, head - marcom and optical business, Anritsu India Pvt Ltd, says, "Anritsu has been the leader in handheld portable T&M instruments that are widely used in aviation communication system maintenance and deployment. Vector network analyser (VNA)

Master MS2025B/35B is used for instrument landing system cables measurement. VNA Master has a dedicated application (vector voltmeter) for the avionics industry."

VNA Master MS2025B/MS2027C series and Site Master S331E/S361E are used by airplane companies for cable verification after installation in aircraft and during maintenance. These instruments are battery operated and suitable for field applications. These can be taken to antenna or air traffic control sites for various other measurements.

The instruments follow MIL standard for field applications

Fig. 1: How instrument landing system works (Credit: www.quora.com)

